

Exercise Sheet Week 4

Key:

- **Questions marked with a 🍷** are for discussion in your tutorials. Please complete these questions before your tutorial.
- **Questions marked with a 🦋** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🍷 **Exercise 1.** Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ be vectors in $\mathbb{R}^3$. Prove that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$.

🍷 **Exercise 2.** Suppose $\vec{a}, \vec{b}, \vec{c}$ are three vectors in $\mathbb{R}^3$ such that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b} \neq \vec{0}$. Prove that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$.

🎵 **Exercise 3.** For each case, write the vector $\overrightarrow{PQ}$, and find the distance between P and Q

- $P = (-5, -7, 2)$ and $Q = (-2, -3, 4)$
- $P = (1, 5, -3)$ and $Q = (2, 3, 4)$
- $P = (1, 2, -3)$ and $Q = (0, 0, 0)$

🎵 **Exercise 4.** The points A , B , and C have position vectors $\vec{a} = 5\hat{i} - 6\hat{j} + 4\hat{k}$, $\vec{b} = 2\hat{i} + 5\hat{j} - 3\hat{k}$ and $\vec{c} = -\hat{i} + 4\hat{k}$, respectively.

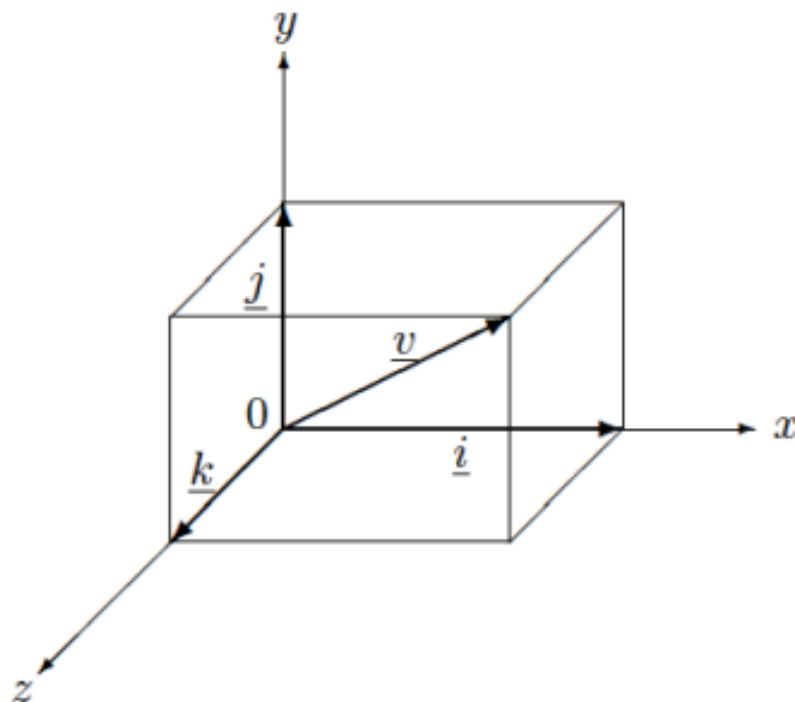
(i) Find

- $5\vec{a} - 2\vec{b} - 3\vec{c}$
- $\overrightarrow{AC} + \overrightarrow{AB}$
- $\vec{a} \cdot \vec{b}$
- θ , where θ is the angle between $\vec{a}$ and $\vec{b}$.
- $\vec{a} \times \vec{c}$
- $(4\vec{a} - 6\vec{b}) \cdot (2\vec{a} + 3\vec{c})$

(ii) Show that $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2\vec{a} \times \vec{b}$

🎵 **Exercise 5.** Prove that the line joining the mid-points of two sides of a triangle is parallel to the third side and of half its length.

🦋 **Exercise 6.** Consider a unit cube whose edges are $\hat{i}$, $\hat{j}$, and $\hat{k}$ with initial points at the origin. The vector $\vec{v}$ is the diagonal of the cube, see the diagram below.



Express $\underline{v}$ in terms of $\hat{i}$, $\hat{j}$ and $\hat{k}$. Hence find the following angles

- (i) the angle between the diagonal of the cube and one of its edges
- (ii) the angle between the diagonal of the cube and the diagonal of one of its faces.